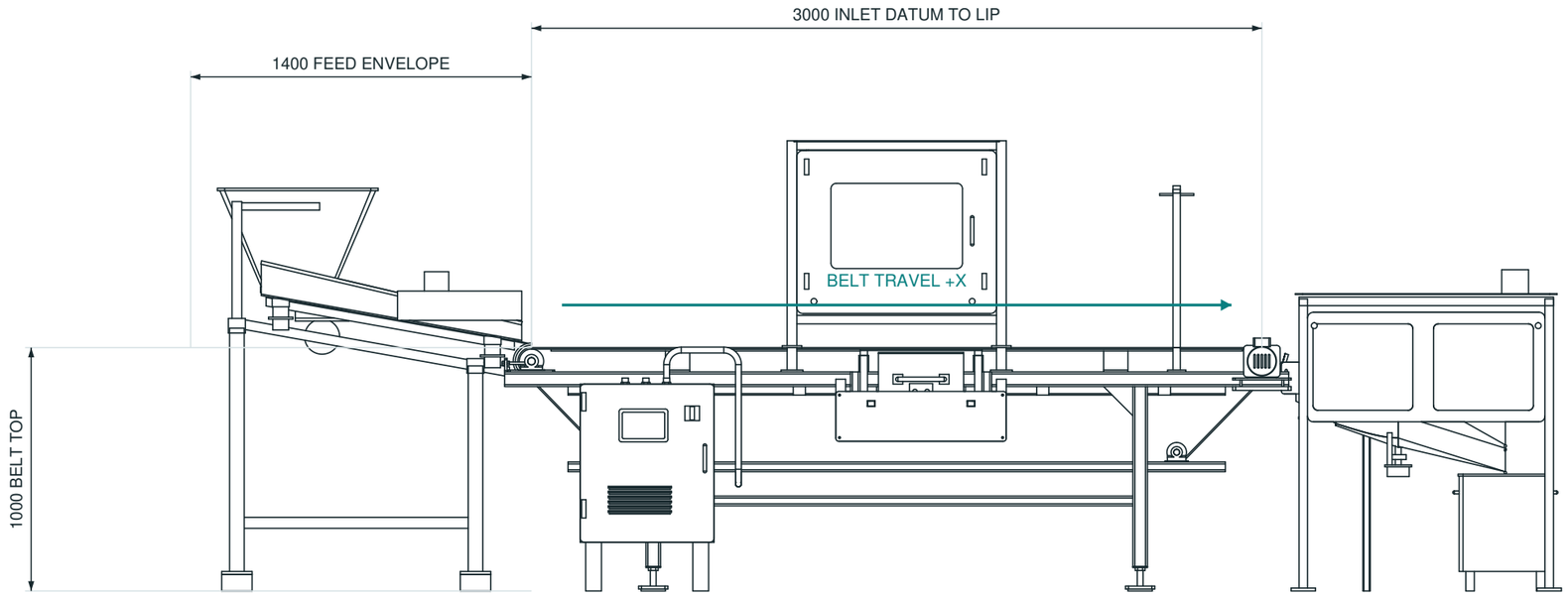
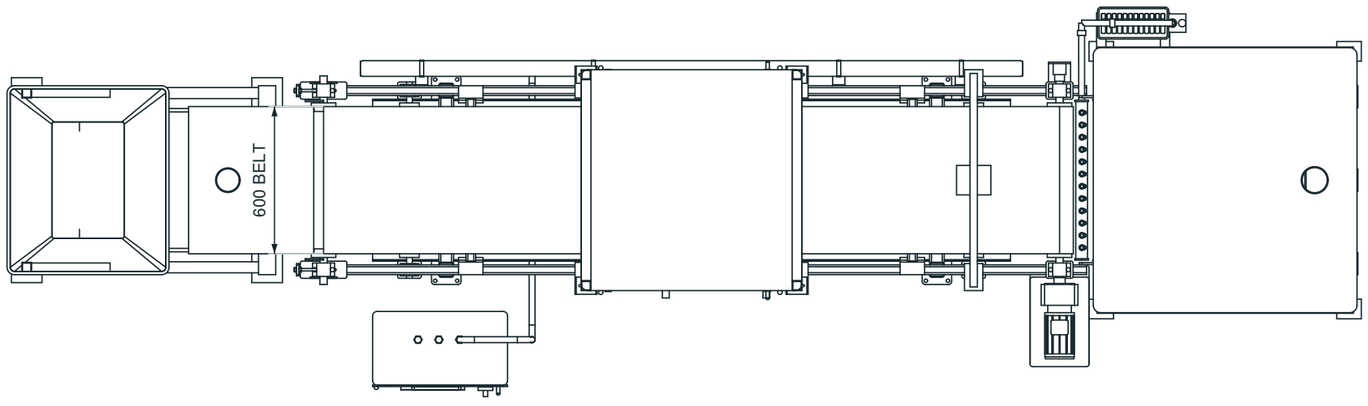
General arrangement

Proposed PCB-rich recovery module / depolluted, liberated, dry screened WEEE / geometry only



SIDE / actual assembly, covers retained / scale per drawing / 1:25

Side scale 1:25; nominal floor z=0. Feed and discharge hardware are concept selections.



PLAN / same solid assembly, orthographic +Z view / 1:25

DESIGN INTENT

One airjet target/balance split; a separate balance-path gate diverts conservative review windows.

COORDINATE DATUMS

x follows belt travel; y is transverse; z is height above floor. Inlet x=0 and lip x=3000 are provisional coordination datums.

Station	x (mm)
RGB centre	1300
Induction centre	1600
Last active datum	1750
Cassette edge	1775
Discharge lip	3000

KEEP THESE DISTINCT

The sensing datum is 25 mm before the enclosure edge. The 1250 mm tracking distance starts at x=1750, not x=1775.

DRAWING STATUS

STEP-derived views show packaging and mechanical intent. Strength, wear, drive sizing, guarding and component selection remain open.

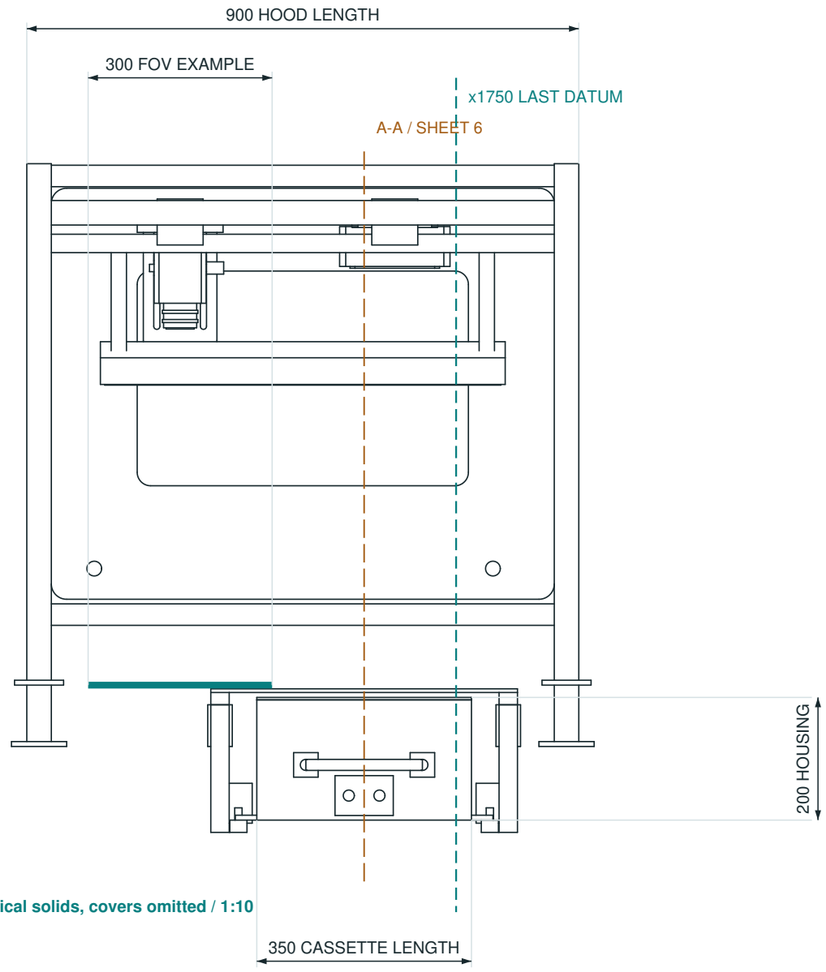
DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

Preliminary study geometry. Proposed inspection targets on sheet 8 are unapproved; no load, sensing or safety rating.

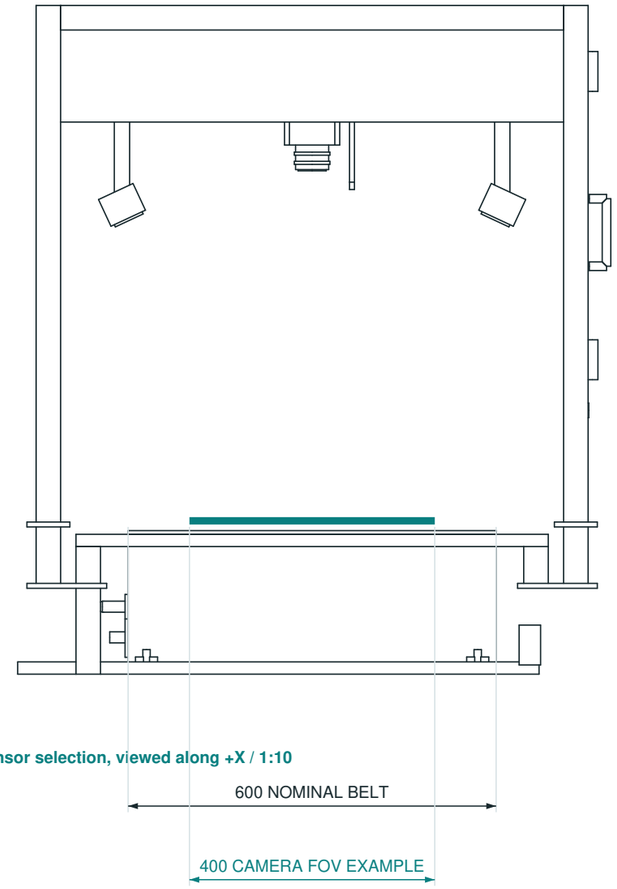
Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Sensing cell and coverage

Adjustable RGB and induction envelopes / optional depth device is unbuilt / opaque hood panels omitted



SENSOR SIDE / selected physical solids, covers omitted / 1:10



TRANSVERSE VIEW / sensor selection, viewed along +X / 1:10

CONDITIONAL CASSETTE REMOVAL

Remove the left access panel, withdraw locating pins and disconnect connectors. Reserve an engineered isolation procedure. This drawing checks clearance only, not safe servicing.

LIFT-OFF IS NOT THIS HOUSING GAP

Selected housing-to-window gap: 8 mm, packaging only. Coil plane, belt/window effects and object shape require calibration.

COVERAGE REQUIRES CALIBRATION

400 mm optical coverage is an example, not full 600 mm belt coverage. The separate 500 mm effective-width calculation is an assumption. Usable sensing width is TBD.

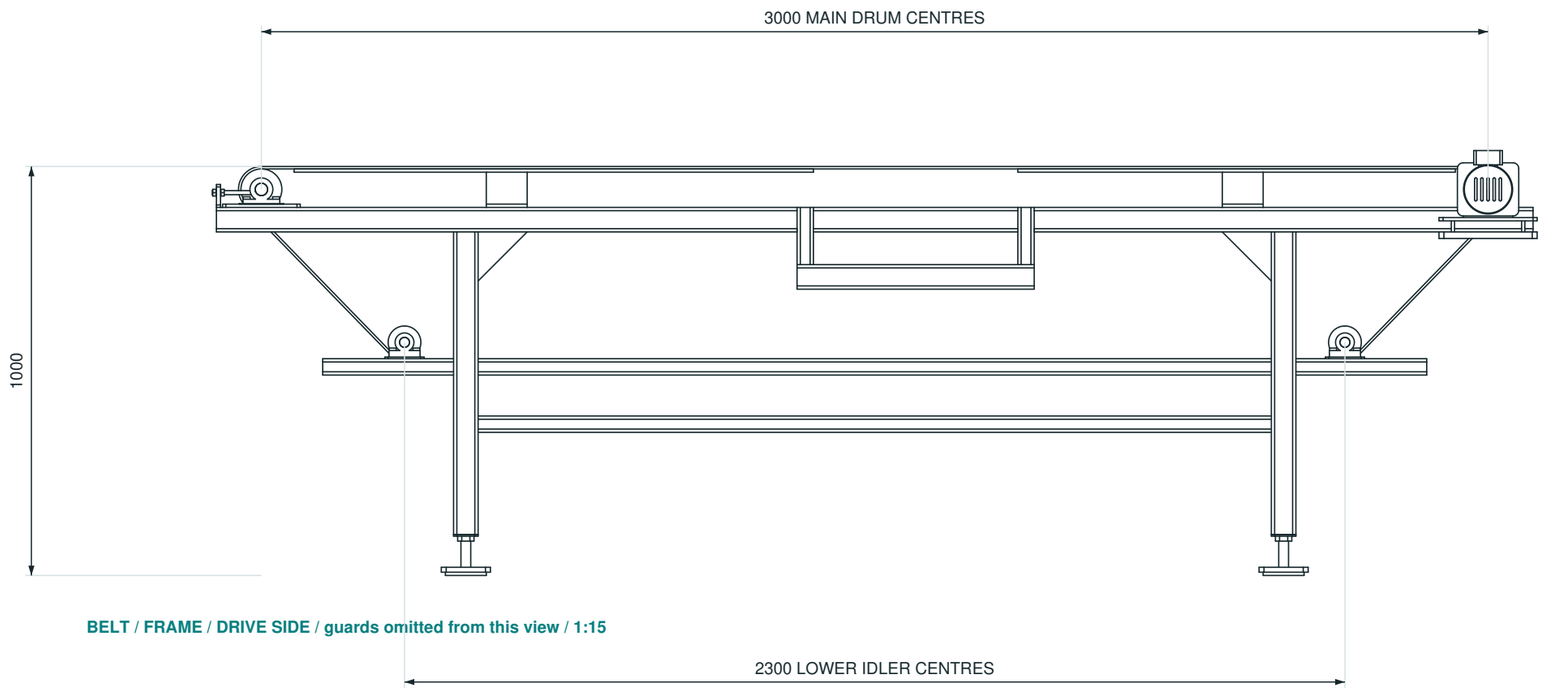
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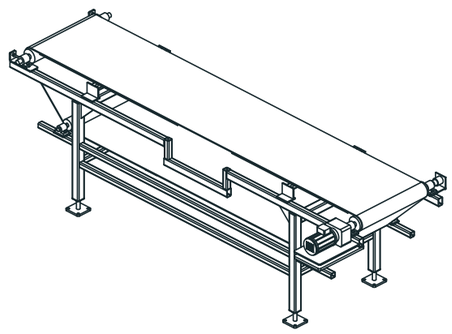
Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Belt path, support and drive

Selected mechanical layout / covers removed for visibility / sizes are not strength-checked



BELT / FRAME / DRIVE SIDE / guards omitted from this view / 1:15



BASE ISOMETRIC / actual CAD solids / no manufacturing scale

BELT PATH SELECTION

Main drums: diameter 100, centres z=944. Belt thickness 6. Lower idlers: diameter 80, centres z=570. Lower belt outer bottom z=524.

SENSOR KEEP-OUT

The lowered return run clears the induction housing at z=780..980. The nonconductive window spans x=1350..1850; placeholder thickness 6 requires deflection and wear tests.

DRIVE AND TENSION

Motor, gearbox, bearings, shafts, tension screws and mounts are modeled packaging selections. Motor torque, belt tension, tracking authority and bearing life are not calculated.

RELEASE INPUTS

Define maximum piece mass, impact/load cases, belt construction and joint, roller crown, shaft/bearing fits, drive duty and restart torque before detailing fabrication.

PROTECTION

Guard and access-panel geometry is a concept allowance. Nip clearances, interlocks, isolation, emergency functions and dust controls require a competent machinery review.

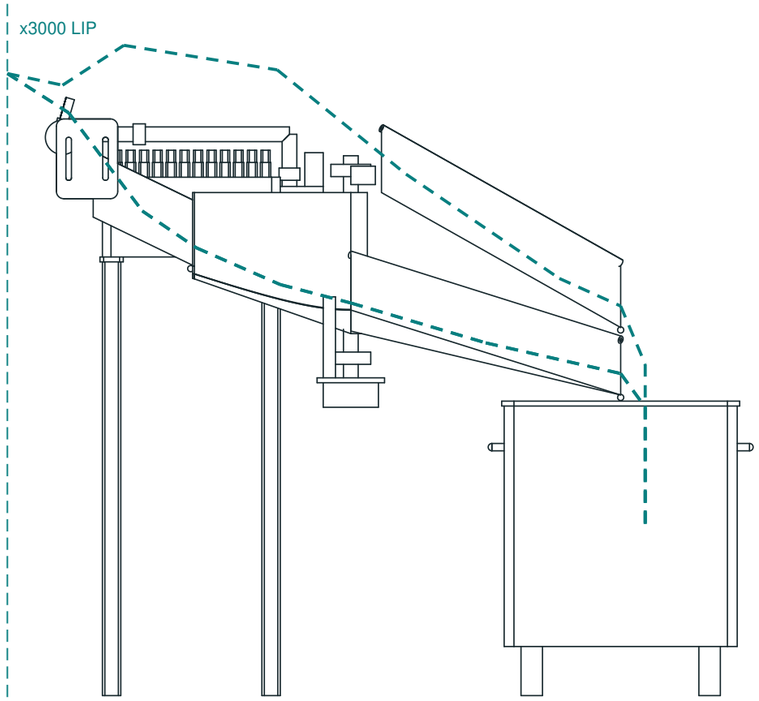
DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

Preliminary study geometry. Proposed inspection targets on sheet 8 are unapproved; no load, sensing or safety rating.

Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Full-width capture and review routing

Repaired collection geometry / one proposed airjet split plus downstream balance-window gate / no measured flow claim

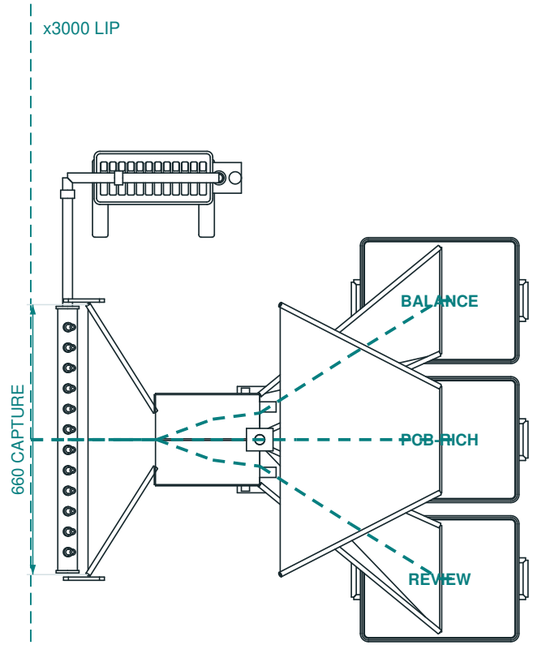


DISCHARGE SIDE / selected solids, protective shell omitted / 1:10

Dashed balance/review paths are alternatives; all paths are intent, not measured trajectories.

TARGET CAPTURE / TRANSVERSE CONVERGENCE

The 660 mm concentrate inlet spans the nominal 600 mm belt and converges to a 260 mm outlet. Sidewalls provide a physical narrowing path; they do not prove friction, rebound, bridging or recovery. Jet impulse and delay remain test inputs.



DISCHARGE PLAN / physical chute and collection layout / 1:15

BALANCE CAPTURE / GROUP GATE

A 660 mm catch converges to the 136 mm gate approach. The upstream-pointing blade is shown in transition position: no feed permitted. Inhibit feed, drain, command and verify gate position before resuming. See sheet 10.

ROUTING BOUNDARY

Review routing may include neighbouring pieces and cannot recover an unknown already ejected into concentrate. Chute bounce prevents exact item-time assumptions. Feed inhibit and a normal review bin are not certified safety functions or battery containment.

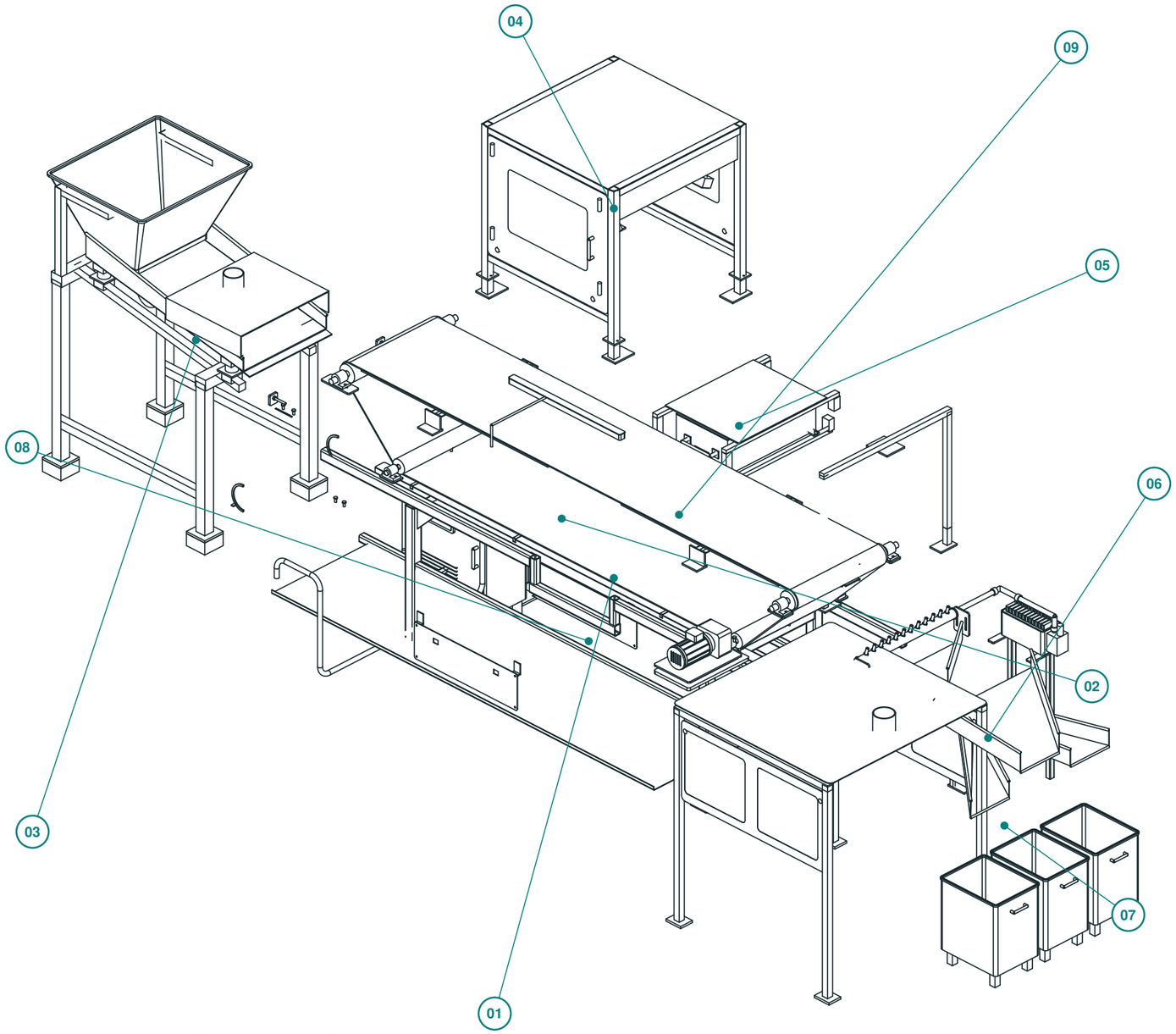
DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

Preliminary study geometry. Proposed inspection targets on sheet 8 are unapproved; no load, sensing or safety rating.

Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Assembly groups and concept BOM

Exploded arrangement from the same STEP solids / group translation for visibility only



EXPLODED ISOMETRIC / arbitrary separation / not to scale

No.	Assembly group	Parts
01	Base, deck and belt	63
02	Drive / bearings / fixings	54
03	Feeder / spreader	61
04	Hood, light, camera	81
05	Sensor and window	27
06	Airjet and chutes	89
07	Gate and bins	41
08	Guards and cable	45
09	Controls / research	33

494 modeled part instances; quantities from manifest.

BOM INTERPRETATION

Counts describe modeled solid part instances, not purchased assemblies. Sensor, actuator and cabinet internals are envelope models. Fixings shown are illustrative; final fastener schedules remain open.

SOURCE TRACEABILITY

The part manifest retains names, categories, bounding boxes and solid volumes. The parametric CadQuery source is the editable design definition. STEP is the checked exchange solid assembly.

NOT A PROCUREMENT RELEASE

Material grades, finishes, fabrication details, approved vendors, quantities of unmodeled internals, torque values and acceptance tolerances are not specified.

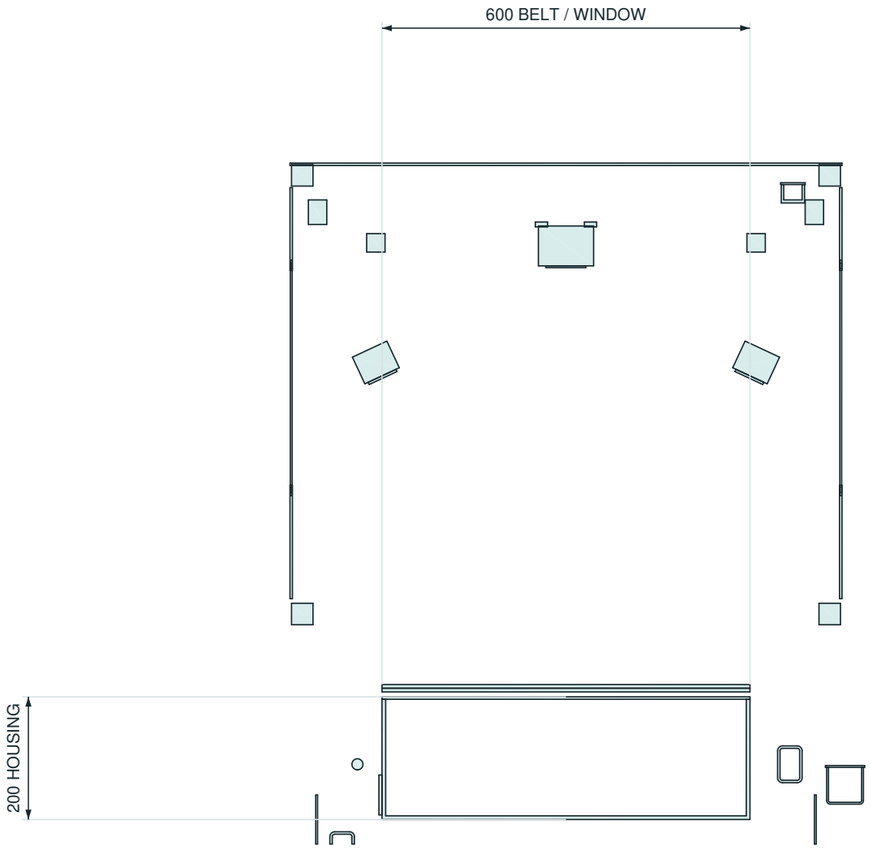
DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

Preliminary study geometry. Proposed inspection targets on sheet 8 are unapproved; no load, sensing or safety rating.

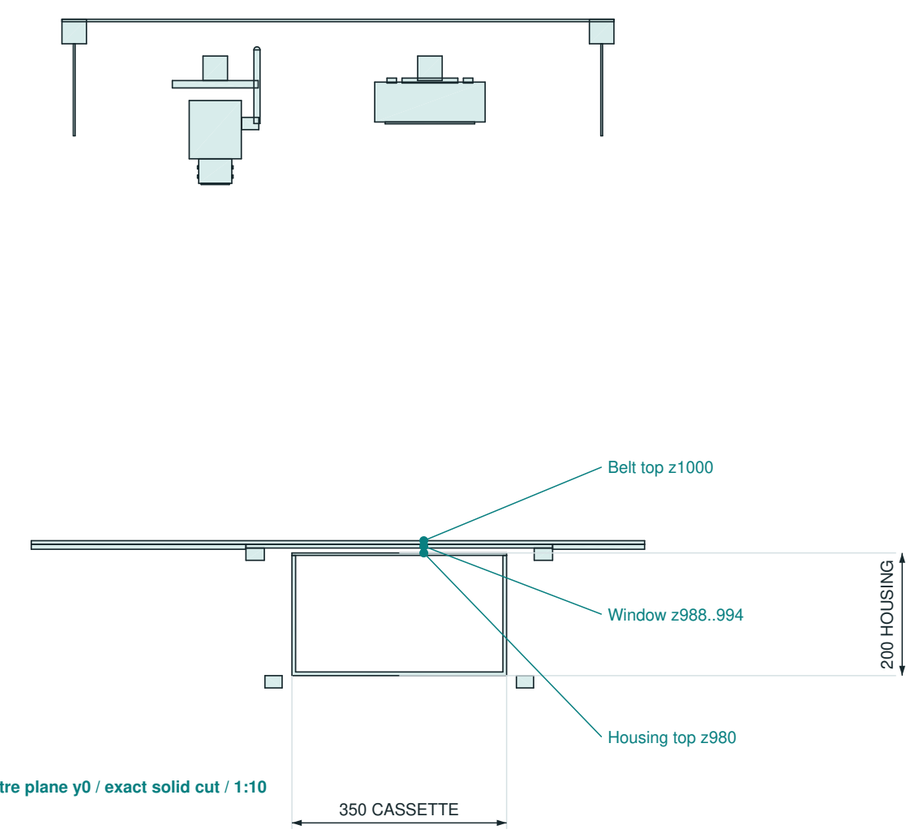
Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Actual solid sections / sensing interfaces

A-A at x=1600 / B-B at y=0 / tinted regions are exact intersections with exported solids



A-A / transverse plane x1600 / exact solid cut / 1:10



B-B / longitudinal centre plane y0 / exact solid cut / 1:10

WHAT THE SECTION PROVES

Actual solid intersections expose the belt, support window, cassette shell and mounting interfaces at these planes. The optional depth envelope is unbuilt. A hollow enclosure is not a coil design or validated sensor field.

KEEP THE MEASUREMENT REGION NONMETALLIC

No steel slider bed is proposed in the reserved sensing region. Nearby steel, wiring, belt constituents and return geometry still require electromagnetic baseline and interference tests.

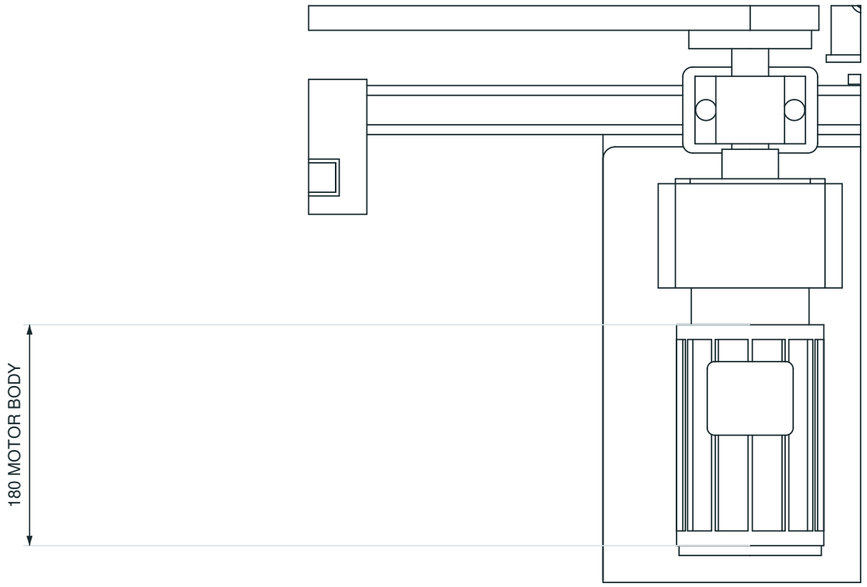
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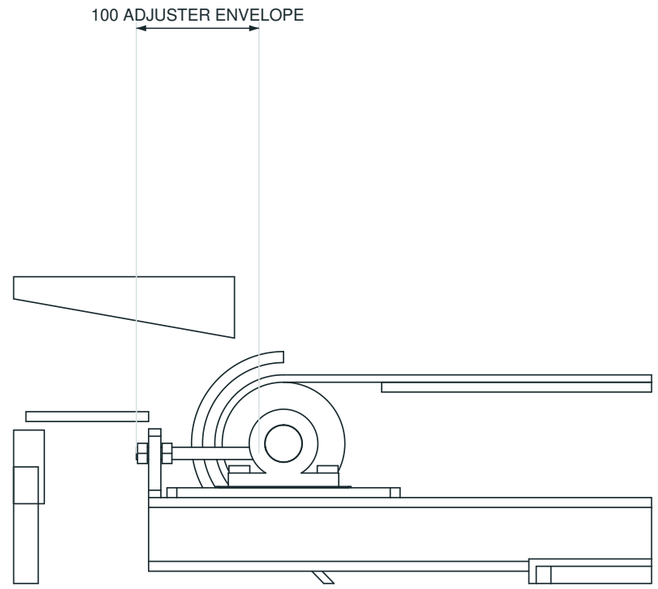
Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Drive, tensioning and control interfaces

Localized views from the same solid assembly / preliminary packaging / no selected manufacturer internals



DRIVE DETAIL / top view / surrounding geometry cropped / 1:5



TAIL DETAIL / side view / surrounding geometry cropped / 1:5

DRIVE + SHAFT ENCODER

The motor, reducer and coupling indicate an installation concept. Encoder feedback is taken at the head shaft; independent belt-slip measurement remains open. Power, reduction, torque, encoder resolution and duty cycle are not selected.

METROLOGY INTERFACE

Datum A: nominal floor z0. B: belt centre plane y0. C: inlet plane x0. Map sensor data to the belt frame; keep camera extrinsics and the final sensing datum in the same measured coordinate system.

TENSION + TRACKING

Slotted tail bearing mounts and threaded-adjuster envelopes allow a study of adjustment. The modeled screw is not a thread specification. Usable stroke, locking, pretension and tracking authority must be engineered and tested.

SERVICE CONNECTIONS

Provide maintainable cable loops, separation from moving parts and a documented disconnect path. Cable fill, bend radii, earthing, ingress, electrical design and air isolation are unresolved.

Proposed inspection targets and release gates

Review assumptions only / values below are proposed prototype metrology targets, not approved manufacturing tolerances

ID	Interface	Proposed target	Verification method	Required release gate
T01	Frame coordination lengths	+/-2 mm	Measure unloaded against datum A/B/C	Structural/mechanical design approval
T02	Head/tail shaft parallelism	0.3 mm over 600 mm	Indicator or laser alignment, unloaded	Belt tracking trial under load
T03	Camera bracket repositioning	+/-1 mm translation	Fiducial recheck after repositioning	Pixel-to-belt calibration and blur test
T04	Cassette mount repositioning	+/-0.5 mm translation	Gauge installed mount reference points	Response/repeatability versus lift-off
T05	Nonmetallic window flatness	0.5 mm unloaded	Straightedge / indicator on test fixture	Loaded deflection, wear and EMC trial
T06	Shaft/bearing fits and runout	TBD - no fit released	Vendor drawing and load calculation	Bearing life, shaft/fit engineering
T07	Nozzles / splitter / chute angles	TBD after sample tests	High-speed measured trajectories	Recovery at fixed purity/throughput
T08	Guards / interlocks / air isolation	TBD after risk review	Competent machinery safety assessment	Validated protection and procedures

NO GENERAL +/- TOLERANCE IS RELEASED

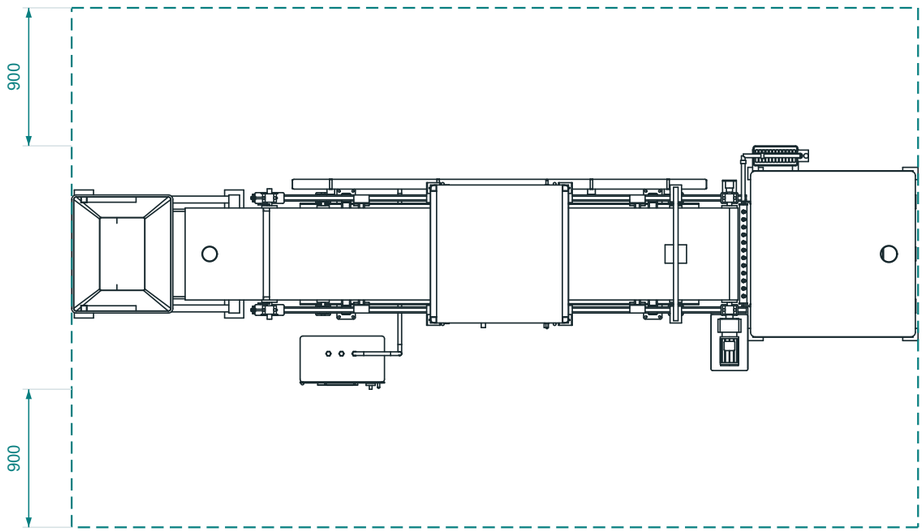
Numeric targets T01-T05 are proposed inspection goals for design discussion. They require tolerance-stack review, process capability, calibrated instruments and acceptance approval before any purchase or fabrication order.

STANDOFFS AND MAXIMUM PIECE SIZE

Camera, depth and induction standoffs remain adjustable. Sieve bands 12.5-25 and 25-50 mm do not specify maximum height, length or mass. Verify the actual sample distribution before freezing clearances or ejection impulse.

Installation envelope and trial gates

Layout reservations, ideal kinematics and tests still needed / no rated machine performance



PLAN / 900 mm side zones from outer equipment / 1:40

SERVICE SPACE IS A RESERVATION

Measure side zones from the outer hardware envelope, including projecting cabinet or drive parts. Door, cassette-withdrawal and bin-exchange sweeps need separate site checks. End access, utilities and actual dust-control connections remain to be designed.

FEED BOUNDARY

Only depolluted, liberated, dry screened WEEE. Run 12.5-25 and 25-50 mm campaigns separately. Exclude batteries, whole devices, hazards, tangles and oversized/heavy pieces; sieve size does not bound length, thickness or mass.

IDEAL KINEMATICS

m/s	mm/frame*	s to lip**
0.5	16.7	2.5
1	33.3	1.25
2	66.7	0.625

* At 30 fps reference. ** 1250 mm / speed. Assumes steady belt speed, no slip or collision. These are calculated values, not latency or throughput evidence.

PROPOSED TEST RAMP

50-200 kg/h is an experimental ramp, not rated capacity. Begin slow and sparse at 0.5 m/s; 1 m/s is the calculation reference and 2 m/s is an unvalidated study case.

MEASURE BEFORE CLAIMING

Calibrated sensor coverage and lift-off; presentation/overlap; synchronized tracking/latency; ejection scatter and neighbours; chute dwell and gate feedback; balanced mass, purity, recovery and actual buyer value.

SEPARATE EXISTING EVIDENCE

The recorded steel/copper YOLO demo remains a separate artifact. It is not PCB inference, physical recovery or evidence of this concept's profitability.

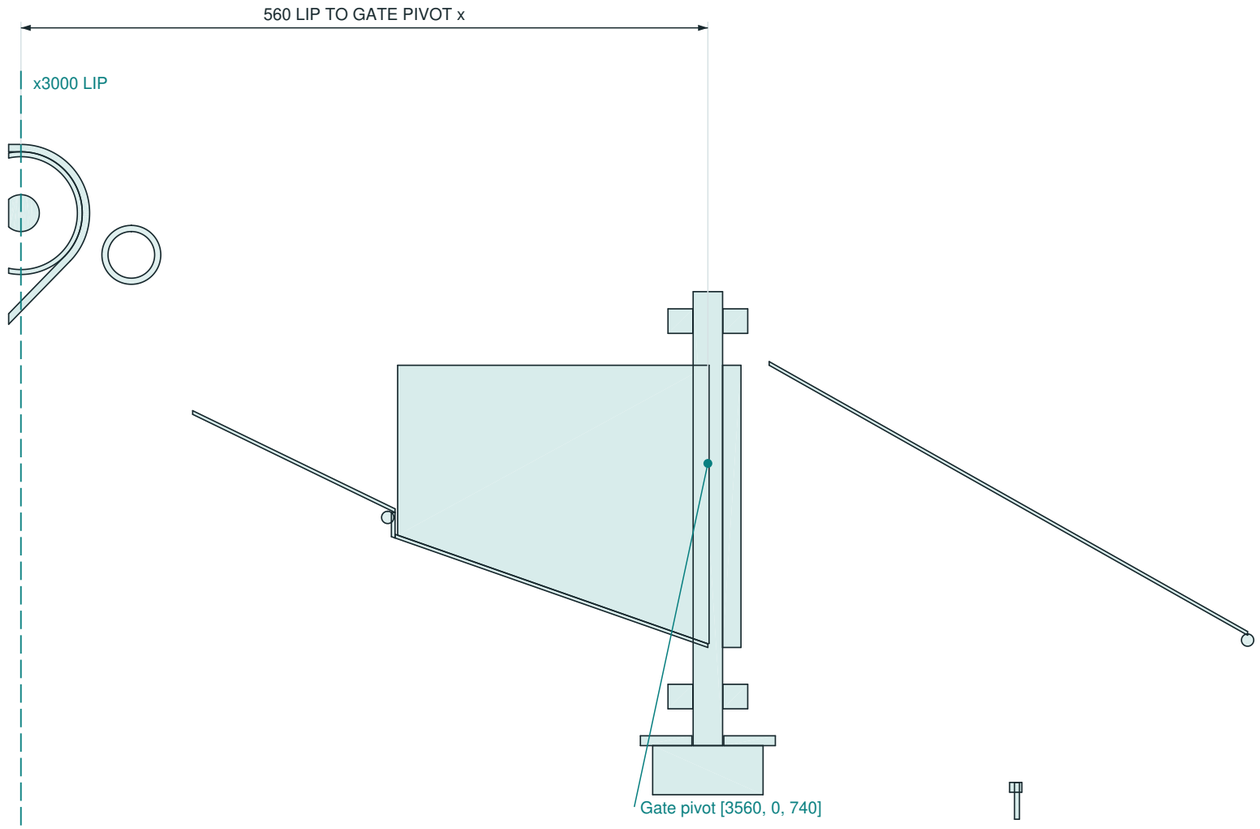
DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

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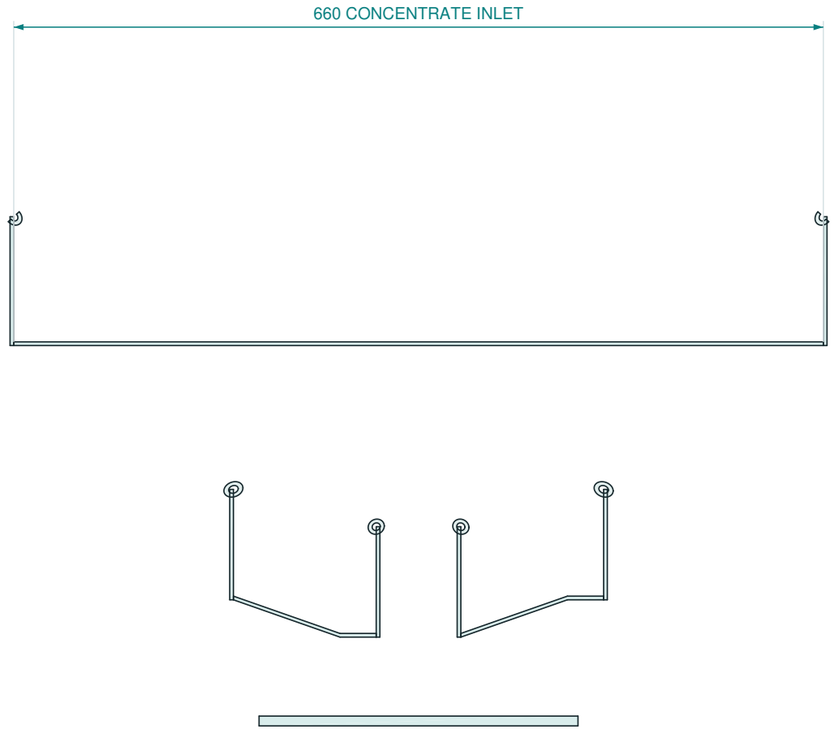
Source: conveyor_assembly.step | SHA-256 80c159dc06435c00 | 494 part instances | basis 2026-10-14

Actual discharge sections / capture repair

C-C at y=0 and D-D at x=3610 / exact solid cuts / geometric coverage does not establish material-flow performance



C-C / y0 / flow-zone crop, exact solid cut / 1:5



D-D / x3610 / concentrate inlet solid cut / 1:5

CAPTURE AND CONVERGENCE

Both 660 mm inlets span the 600 mm belt. Concentrate narrows to 260 mm at x4000. Balance converges to a 136 mm approach before the upstream-pointing gate blade.

GEOMETRY BOUNDARY

Width checks establish only modeled capture and guide surfaces. The 30 mm nominal allowance per belt edge is not a validated trajectory-scatter margin.

BALANCE GATE / CONTROL HOLD

The gate serves balance only and cannot recover already-ejected concentrate. Inhibit feed, verify drain time, command and confirm position, then resume. Transition geometry is not an operating state.

FLOW AND HARDWARE HOLD

Gate seats show nominal contact, not verified closure. Test fragment limits, bounce, jams, leakage and residence time. Actuator, stops, seals, pneumatics and safeguarding remain unresolved.

DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE

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